

Sunny Productivity – ICT171

Assignment 3

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Global IP address: 13.60.14.37

DNS Entry: <https://sunnyproductivity.live/>

Video explainer: <https://youtu.be/KGcnHdkub7s>

GitHub repository: <https://github.com/ieakia/sunny-productivity>

Used Amazon Web Services EC2 with a domain name from 'name.com.'
(sunnyproductivity.live)

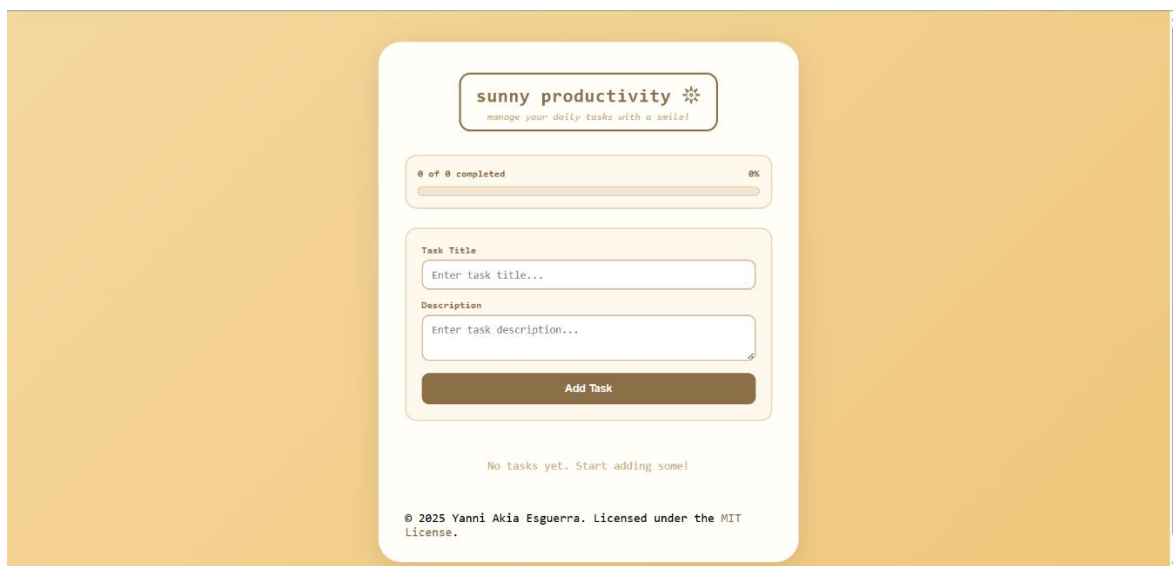
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Project Overview

This project is a task management web application deployed on AWS EC2 using IaaS. The server runs Ubuntu 24.04 with the Apache web server.

‘Sunny Productivity’ is a personal task management tool that aims to help users manage their daily tasks with a user-friendly interface and visual progress tracking.



Technologies Used

- Platform: AWS EC2
- Operating System: Ubuntu 24.04
- Instance type: t3.micro
- Web server: Apache, manually configured/installed
- Front end: HTML, CSS, JavaScript
- Security and DNS: Let's Encrypt (Certbot) (SSL), Name.com (DNS Provider)

Set up Documentation

Launching AWS EC2 Instance

Step 1: Launching instance

1. After signing into the AWS Console, launch an EC2 instance.
2. Choose an Amazon Machine Image (AMI) – select Ubuntu Server 24.04 LTS
3. Pick an instance type: t3.micro (free tier)
4. Create a new key pair; name it and save it in the .pem format. Download and save it securely.
5. Configure the security group
6. Launch instance and note down the Public IP address.

Step 2: Connecting to the instance

1. Access the instance from your terminal (Windows PowerShell) via ssh.

Command used: `ssh -i "todolist.pem" ubuntu@13.60.14.37`

```
PS C:\Users\Yanni Akia\onedrive\documents> ssh -i "todolist.pem" ubuntu@13.60.14.37
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1015-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Nov 15 22:37:11 UTC 2025

System load:  0.0               Temperature:   -273.1 C
Usage of /:   33.4% of 8.65GB   Processes:    121
Memory usage: 63%              Users logged in: 0
Swap usage:   0%               IPv4 address for ens5: 172.31.25.164

 * Ubuntu Pro delivers the most comprehensive open source security and
   compliance features.

https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
```

2. Update system packages

Commands used:

- `sudo apt update`
- `sudo apt upgrade -y`

3. Install Apache

Commands used:

```
sudo apt install apache2
```

```
ubuntu@ip-172-31-25-164:~$ sudo apt install apache2
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
apache2 is already the newest version (2.4.58-1ubuntu8.8).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
ubuntu@ip-172-31-25-164:~$ |
```

Start and enable Apache.

Ensure it is running: `sudo systemctl status apache2`

```
ubuntu@ip-172-31-25-164:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Thu 2025-11-13 13:57:55 UTC; 5min ago
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 2149 (apache2)
    Tasks: 55 (limit: 1008)
   Memory: 5.4M (peak: 5.7M)
      CPU: 55ms
   CGroup: /system.slice/apache2.service
           └─2149 /usr/sbin/apache2 -k start
             └─2151 /usr/sbin/apache2 -k start
               └─2153 /usr/sbin/apache2 -k start

Nov 13 13:57:55 ip-172-31-25-164 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Nov 13 13:57:55 ip-172-31-25-164 systemd[1]: Started apache2.service - The Apache HTTP Server.
ubuntu@ip-172-31-25-164:~$
```

Configure DNS to point to the Public IP address

I used name.com to obtain a free domain name and pointed it to my website's public IP address.

The screenshot displays the Name.com dashboard for a domain. On the left, there are 'Quick Actions' (Automatic Renewal, Whols Privacy, Transfer Locked) and 'Domain Actions' (Manage DNS Records, Manage Nameservers, URL Forwarding, Email Forwarding). The main area is titled 'Manage DNS' and includes a note about DNS changes taking up to 48 hours to apply. Below this, there are buttons for 'Show DNS Templates', 'Delete Selected Records', and 'EXPORT DNS RECORDS (CSV)'. A table lists existing DNS records:

	TYPE	HOST	ANSWER	TTL	PRIO	ACTIONS
☐	A	sunnyproductivity.live	13.60.14.37	300	N/A	Edit Delete
						CREATED: 2025-11-16
☐	A	www.sunnyproductivity.live	13.60.14.37	300	N/A	Edit Delete
						CREATED: 2025-11-16

Below the table is a form to add a new record:

TYPE	HOST	ANSWER	TTL	PRIO	ACTIONS
A	sunnyproductivity.live		300	N/A	Add Record

Set up SSL with Let's Encrypt

sudo apt install certbot python3-certbot-apache -y

```
ubuntu@ip-172-31-25-164:~$ sudo apt install certbot python3-certbot-apache -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
certbot is already the newest version (2.9.0-1).
python3-certbot-apache is already the newest version (2.9.0-1).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
ubuntu@ip-172-31-25-164:~$ |
```

sudo certbot --apache -d sunnyproductivity.live -d

```
ubuntu@ip-172-31-25-164:~$ sudo certbot --apache -d sunnyproductivity.live -d www.sunnyproductivity.live
Saving debug log to /var/log/letsencrypt/letsencrypt.log
Requesting a certificate for sunnyproductivity.live and www.sunnyproductivity.live

Successfully received certificate.
Certificate is saved at: /etc/letsencrypt/live/sunnyproductivity.live/fullchain.pem
Key is saved at: /etc/letsencrypt/live/sunnyproductivity.live/privkey.pem
This certificate expires on 2026-02-14.
These files will be updated when the certificate renews.
Certbot has set up a scheduled task to automatically renew this certificate in the background.

Deploying certificate
Successfully deployed certificate for sunnyproductivity.live to /etc/apache2/sites-available/000-default-le-ssl.conf

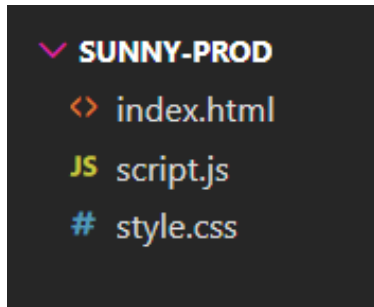
We were unable to find a vhost with a ServerName or Address of www.sunnyproductivity.live.
Which virtual host would you like to choose?
-----
1: 000-default.conf | | | Enabled
2: 000-default-le-ssl.conf | sunnyproductivity.live | HTTPS | Enabled
-----
Select the appropriate number [1-2] then [enter] (press 'c' to cancel): 2
Successfully deployed certificate for www.sunnyproductivity.live to /etc/apache2/sites-available/000-default-le-ssl.conf
Congratulations! You have successfully enabled HTTPS on https://sunnyproductivity.live and https://www.sunnyproductivity.live
```

Check the certificate details:

```
ubuntu@ip-172-31-25-164:~$ sudo certbot certificates
Saving debug log to /var/log/letsencrypt/letsencrypt.log

-----
Found the following certs:
Certificate Name: sunnyproductivity.live
Serial Number: 6205afb1dcccb7040bfeb9e4e83bd6313bcc
Key Type: ECDSA
Domains: sunnyproductivity.live www.sunnyproductivity.live
Expiry Date: 2026-02-14 12:40:33+00:00 (VALID: 89 days)
Certificate Path: /etc/letsencrypt/live/sunnyproductivity.live/fullchain.pem
Private Key Path: /etc/letsencrypt/live/sunnyproductivity.live/privkey.pem
-----
ubuntu@ip-172-31-25-164:~$ |
```

Create the website code using VS Code



Deploy these files to /var/www/html/ by using your local machine to upload them to your instance.

`scp -i todolist.pem index.html script.js style.css ubuntu@13.60.14.37:/home/ubuntu`

```
PS C:\Users\Yanni Akia\onedrive\documents> scp -i todolist.pem index.html script.js style.css ubuntu@13.60.14.37:/home/ubuntu/
index.html      100% 2421    0.9KB/s   00:02
script.js       100% 4644    3.3KB/s   00:01
style.css       100% 6089   28.6KB/s   00:00
PS C:\Users\Yanni Akia\onedrive\documents> |
```

Next, log in to the instance from your terminal, then do the following:

```
ubuntu@ip-172-31-25-164:~$ sudo cp ~/index.html ~/style.css ~/script.js /var/www/html/
ubuntu@ip-172-31-25-164:~$ sudo chmod 644 /var/www/html/index.html /var/www/html/style.css /var/www/html/script.js
ubuntu@ip-172-31-25-164:~$ |
```

Scripting Documentation

One of the main features included in the “script.js” file is a custom modal confirmation system. This replaces the default browser confirm popup with a custom modal popup for a more consistent design and user experience.

```

function deleteTodo(index) {
  deleteIndex = index;
  document.getElementById('modal-overlay').classList.add('active');
}

function closeModal() {
  document.getElementById('modal-overlay').classList.remove('active');
  deleteIndex = -1;
}

function confirmDelete() {
  if (deleteIndex >= 0) {
    todos.splice(deleteIndex, 1);
    if (editingIndex === deleteIndex) {
      cancelEdit();
    }
    renderTodos();
  }
  closeModal();
}

```



This also allows users to exit the pop-up by simply clicking outside the area:


```

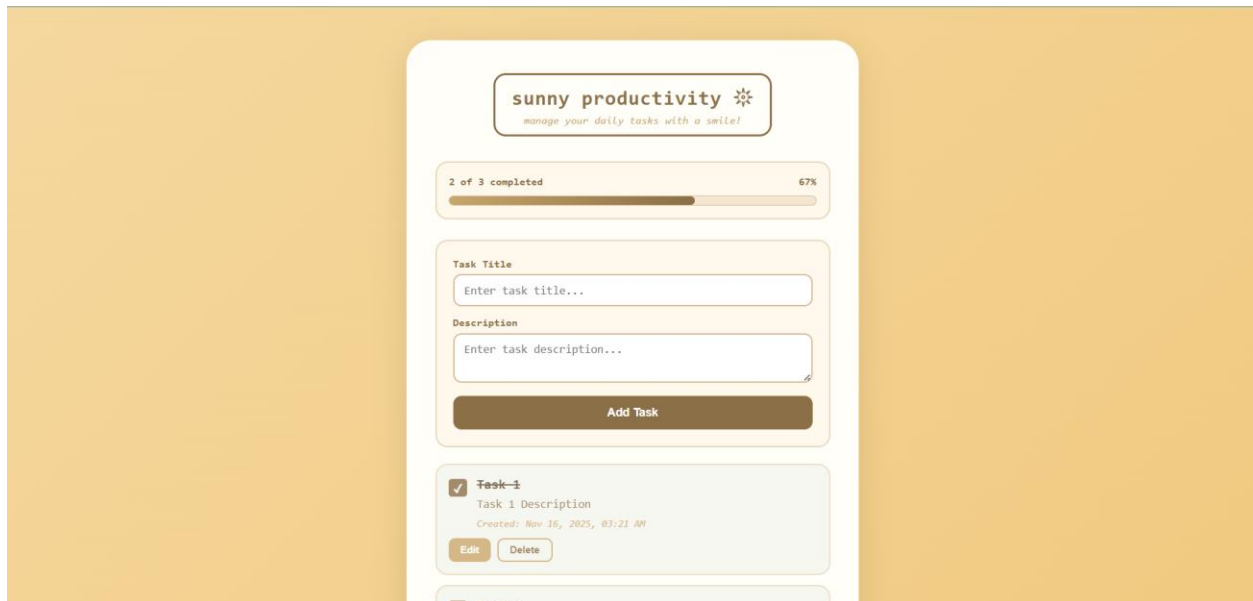
document.getElementById('modal-overlay').addEventListener('click', (e) => {
  if (e.target.id === 'modal-overlay') {
    closeModal();
  }
});

document.getElementById('todo-title').addEventListener('keypress', (e) => {
  if (e.key === 'Enter') {
    addTodo();
  }
});

```

Website Features

- Single-page task manager
- Warm, yellow-themed UI
- Create, Edit, and Delete tasks
- Visual progress tracker with a progress bar
- Interactive features with hover effects over buttons/links
- Mobile compatibility



03:23 ★



sunny productivity ☀

*manage your daily tasks with a
smile!*

1 of 2 completed

50%



Task Title

Description

Add Task



~~Task 1~~

Task 1 description

Created: Nov 16, 2025 at 03:23 AM

Licensing

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